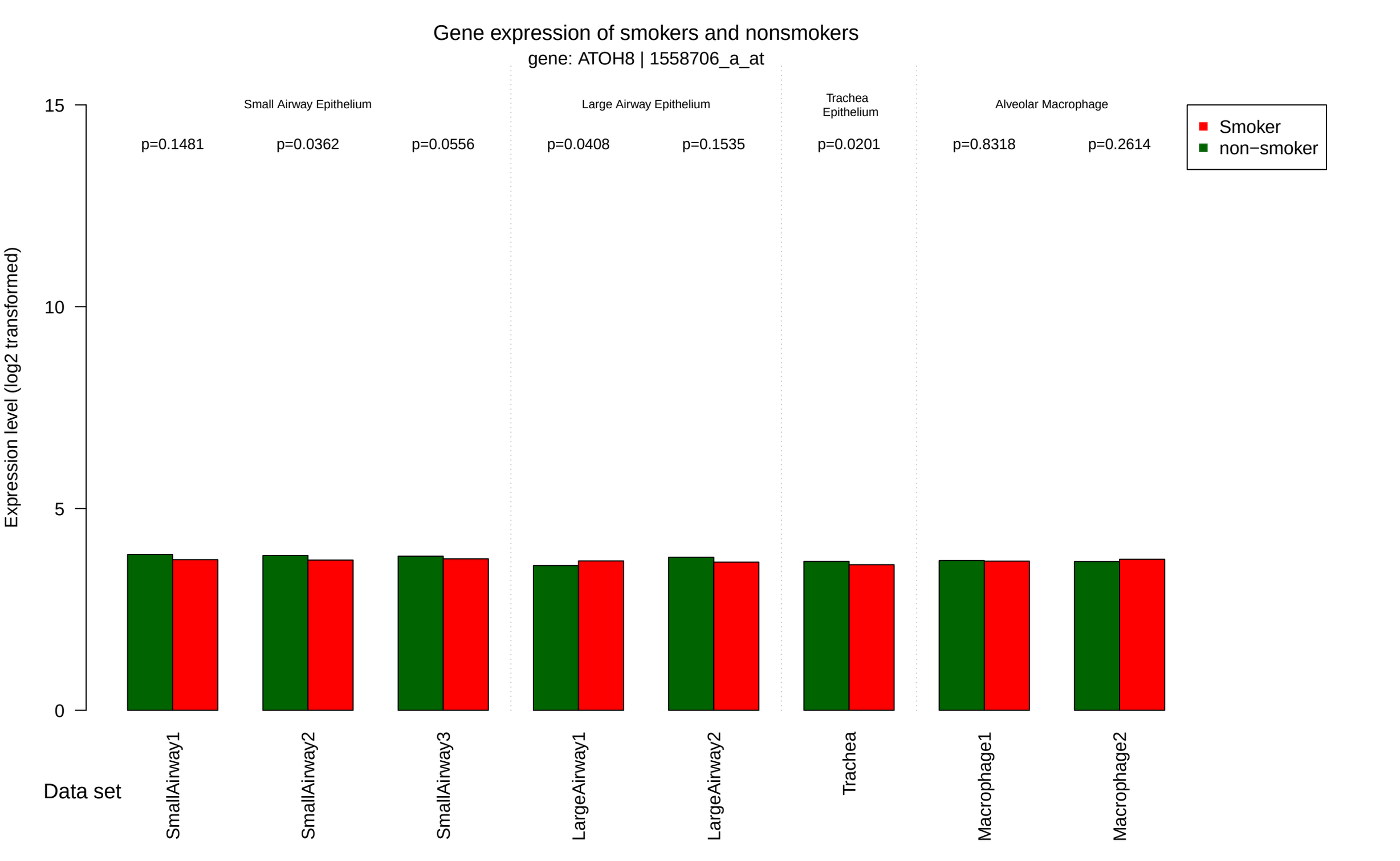
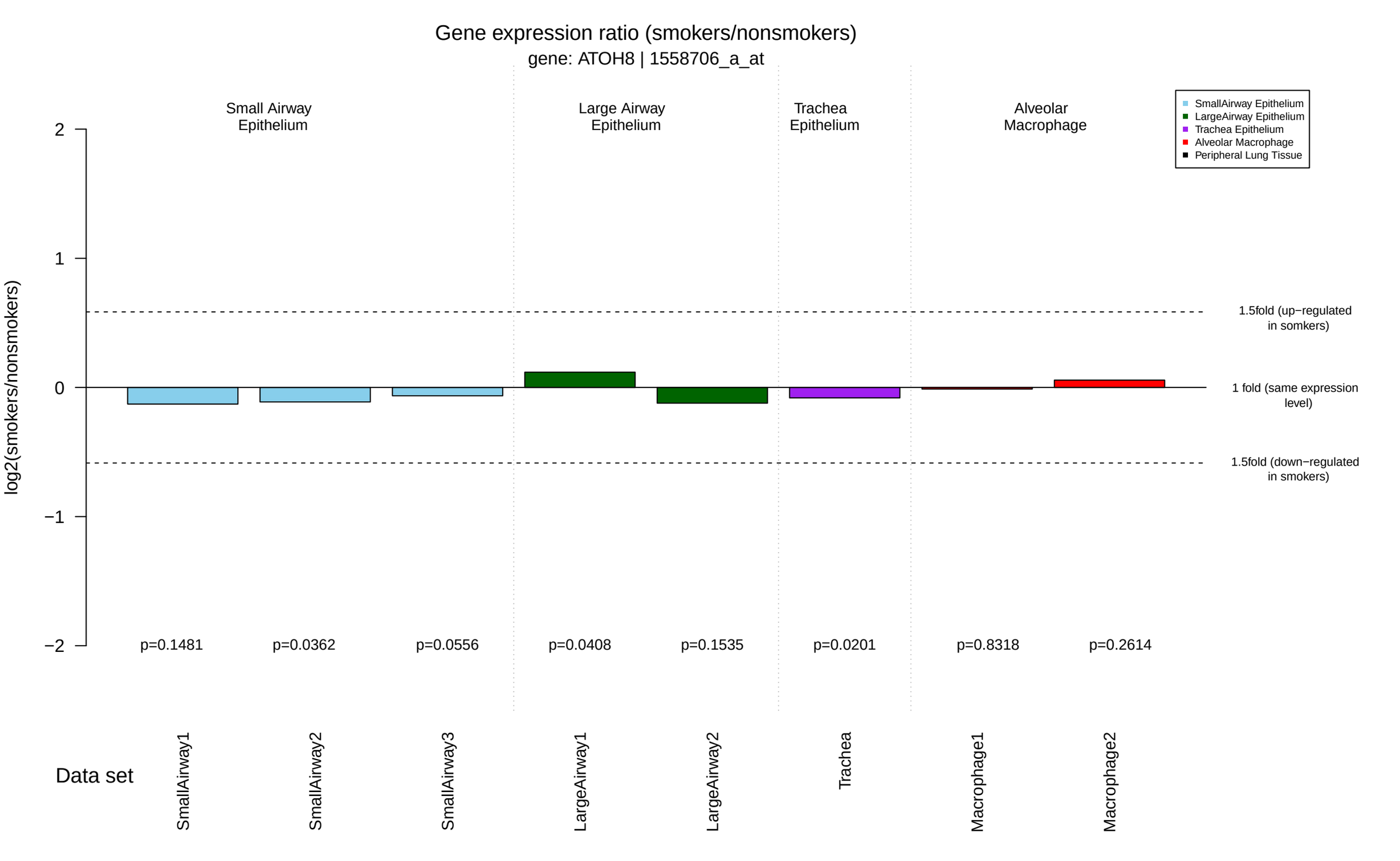
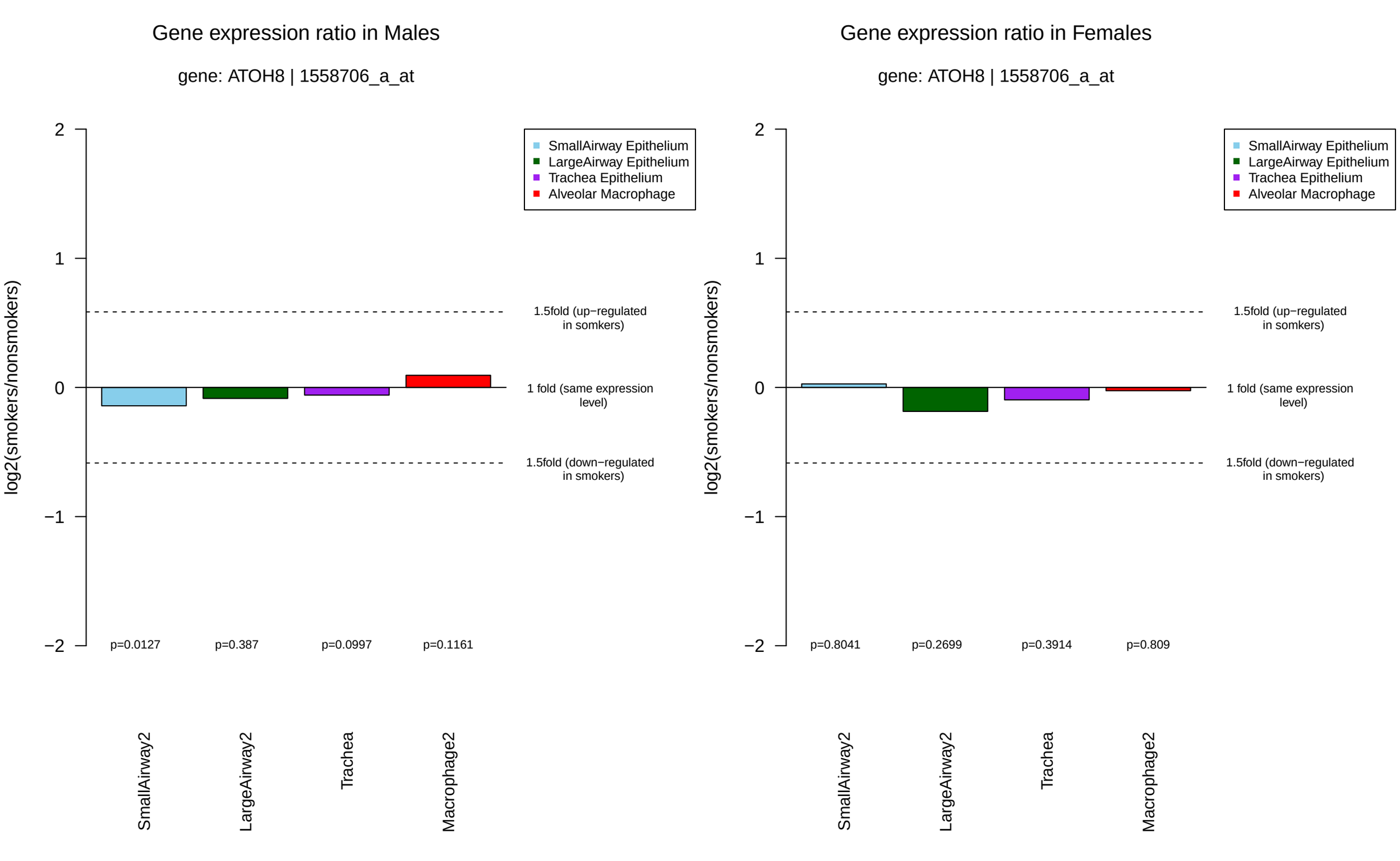
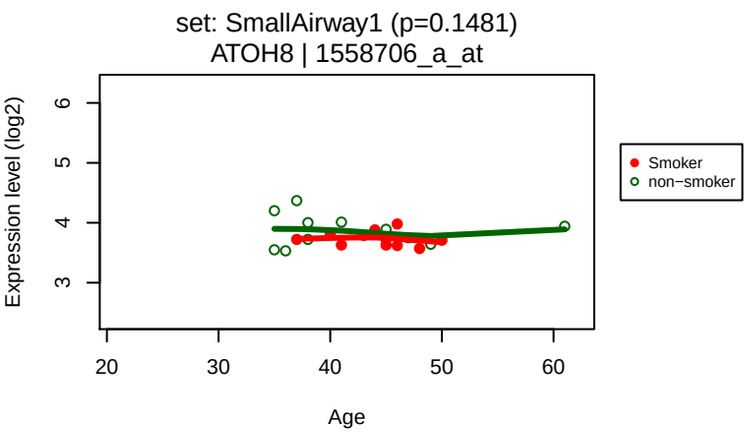
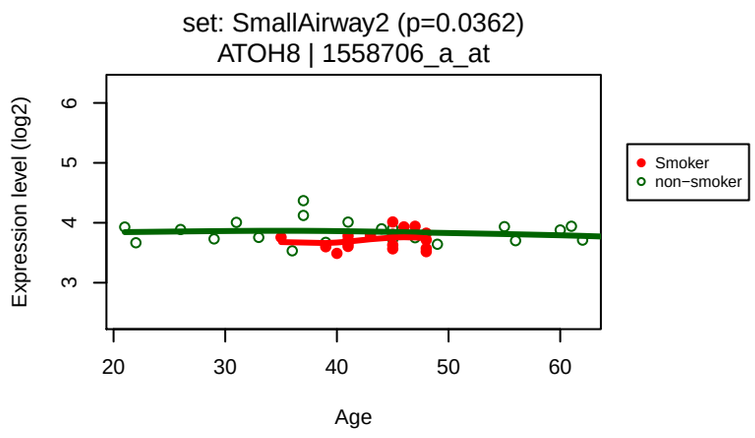
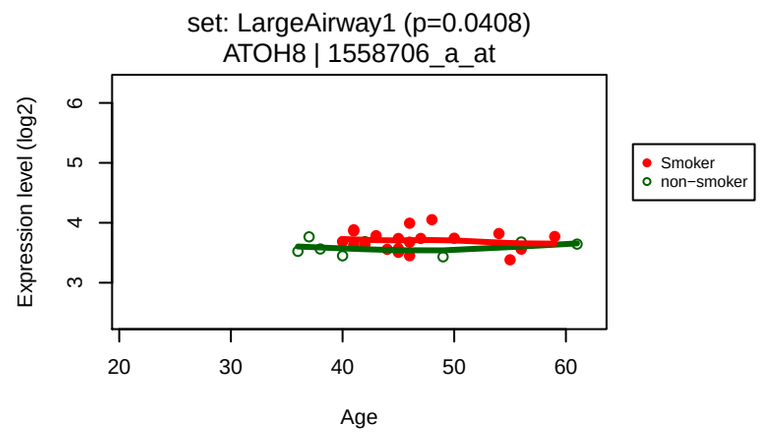

Gene expression of smokers and nonsmokers
gene: ATOH8 | 1558706_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1481
p=0.0362
p=0.0556
p=0.0408
p=0.1535
p=0.0201
p=0.8318
p=0.2614
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

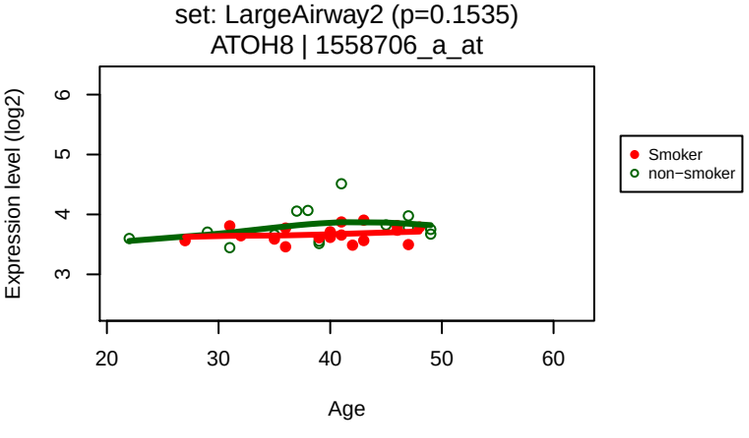

Gene expression ratio (smokers/nonsmokers)
gene: ATOH8 | 1558706_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1481
p=0.0362
p=0.0556
p=0.0408
p=0.1535
p=0.0201
p=0.8318
p=0.2614
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

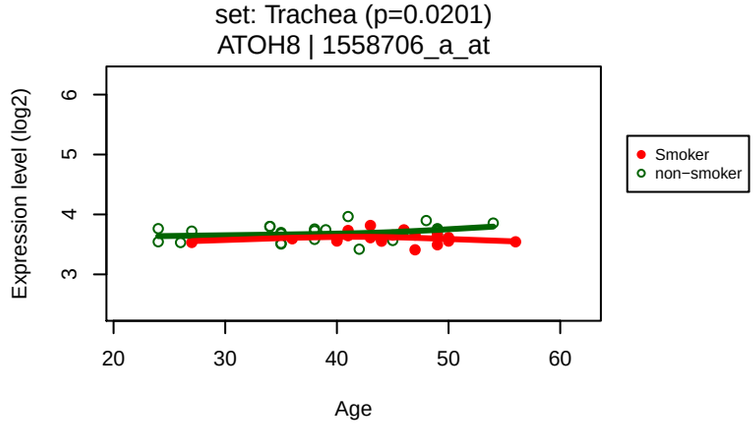

Gene expression ratio in Males
gene: ATOH8 | 1558706_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0127
p=0.387
p=0.0997
p=0.1161
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATOH8 | 1558706_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8041
p=0.2699
p=0.3914
p=0.809
SmallAirway2
LargeAirway2
Trachea
Macrophage2

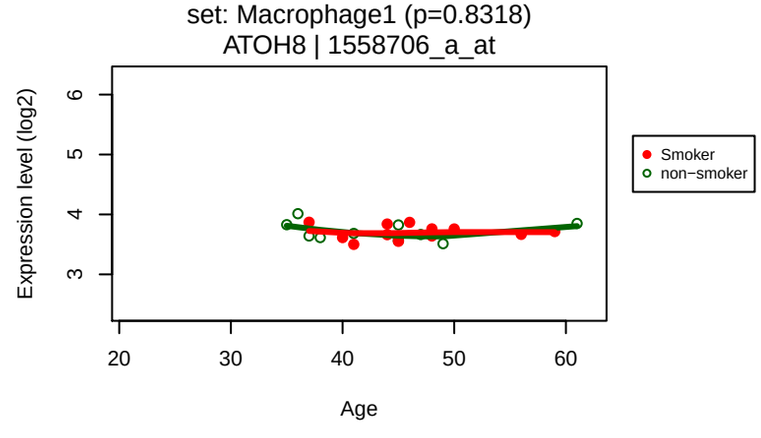

set: SmallAirway1 (p=0.1481)
ATOH8 | 1558706_a_at
Expression level (log2)
Age
Smoker
non−smoker

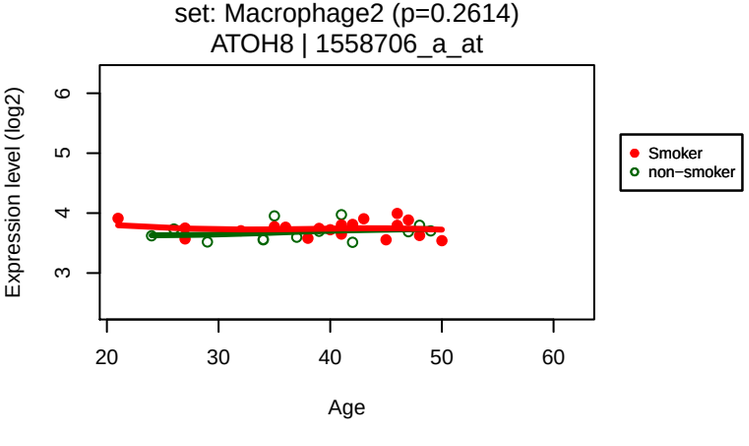

set: SmallAirway2 (p=0.0362)
ATOH8 | 1558706_a_at
Expression level (log2)
Age
Smoker
non−smoker

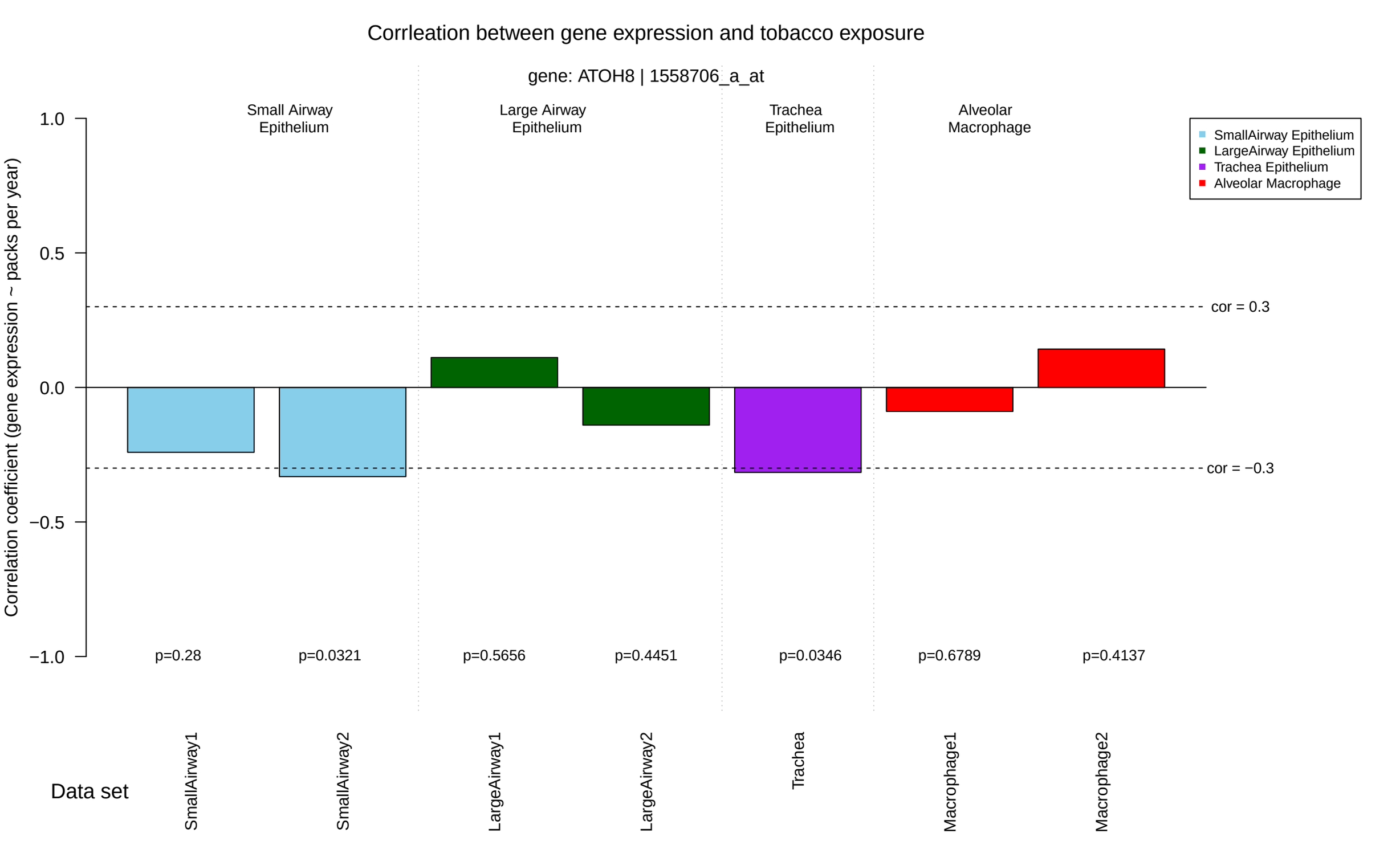

set: LargeAirway1 (p=0.0408)
ATOH8 | 1558706_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1535)
ATOH8 | 1558706_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0201)
ATOH8 | 1558706_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8318)
ATOH8 | 1558706_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2614)
ATOH8 | 1558706_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATOH8 | 1558706_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.28
p=0.0321
p=0.5656
p=0.4451
p=0.0346
p=0.6789
p=0.4137
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2